Worksheet 2.2 — Transformations of Functions

Mathematics: Analysis & Approaches SL (IB) · Unit 2: Functions

The Big Ideas

Every transformed graph is a parent function shifted, stretched, or reflected.

- $y = f(x) + k$ shifts vertically; $y = f(x + k)$ shifts horizontally (opposite the sign).
- $y = af(x)$ stretches vertically by a; $y = f(bx)$ stretches horizontally by $\frac{1}{b}$.
- $y = -f(x)$ reflects in the x-axis; $y = f(-x)$ reflects in the y-axis.

Worked Examples

Example 1 — Describe a transformation

Describe the transformation from $y = x^2$ to $y = (x + 4)^2 - 6$.
Solution. Shift left 4, down 6; new vertex $(-4, -6)$.
Conclusion: left 4, down 6.

Example 2 — Vertical stretch and reflection

Describe $y = -3x^2$ as a transformation of $y = x^2$.
Solution. Factor 3 stretches vertically; negative sign reflects in the x-axis.
Conclusion: vertically stretched by 3, reflected in the x-axis.

Example 3 — Root function shift

Describe the transformation from $y = \sqrt{x}$ to $y = \sqrt{x - 5} + 2$.
Solution. Right 5, up 2.
Conclusion: right 5, up 2.

Example 4 — Horizontal compression

Given $f(x) = x^2$, describe $y = f(3x)$.
Solution. $y = (3x)^2 = 9x^2$ — horizontal compression by $\frac{1}{3}$ (equivalently vertical stretch by 9).
Conclusion: compressed horizontally by $\frac{1}{3}$.

Example 5 — Combining a stretch, shift, and reflection

Describe $y = -2(x - 3)^2 + 5$ as transformations of $y = x^2$, in order.
Solution. Stretch vertically by 2, reflect in the x-axis, then shift right 3 and up 5.
Conclusion: reflected, stretched by 2, shifted right 3 up 5; vertex $(3, 5)$.

Example 6 — Transformation of a root function with compression

Given $f(x) = \sqrt{x}$, describe $y = f(4x) - 3$.

Solution. $y = \sqrt{4x} - 3 = 2\sqrt{x} - 3$ — equivalent to a vertical stretch by 2 combined with a shift down 3.

Conclusion: vertical stretch by 2 (from the horizontal compression), shifted down 3.

Example 7 — Tracking an intercept through transformations

A graph $y = f(x)$ has an x-intercept at $x = 5$. Find the x-intercept of $y = f(2x - 4)$ (factor the input first: $f(2(x - 2))$).

Solution. The transformation is a horizontal compression by $\frac{1}{2}$ then a shift right 2. The original root at $x = 5$ maps: first compress ($x = 2.5$), then shift right 2.

Conclusion: new root at $x = 4.5$.

Example 8 — Application — modelling a shifted signal

A sensor's baseline reading is $y = f(t)$. A second, recalibrated sensor reads 3 units higher and lags behind by 2 seconds. Write its model in terms of f.

Solution. "Lags behind by 2 seconds" means the same reading happens 2 seconds later — a shift right by 2. "3 units higher" is a vertical shift up 3.

Conclusion: the second sensor's model is $y = f(t - 2) + 3$.

Example 9 — Reversing a transformation to find the original function

A graph $y = 2f(x+1) - 4$ has a maximum at $(3, 6)$. Find the coordinates of the corresponding maximum on $y = f(x)$.

Solution. **Step 1:** Undo the vertical stretch/shift: original y-value satisfies $6 = 2y - 4 \Rightarrow y = 5$.

Step 2: Undo the horizontal shift: $x + 1 = 3 \Rightarrow x = 2$.

Conclusion: the maximum on $y = f(x)$ is at $(2, 5)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Describe the transformation from $y = x^2$ to $y = (x-3)^2 + 7$.

Question 2

Describe $y = \frac{1}{2}x^2$ as a transformation of $y = x^2$.

Question 3

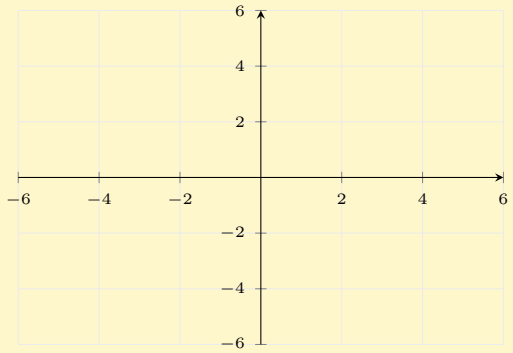
Describe $y = \sqrt{-x}$ as a transformation of $y = \sqrt{x}$.

Question 4

Describe $y = -x^2 - 5$ as two transformations of $y = x^2$.

Question 5

Given $f(x) = x^2$, sketch and describe $y = f(2x)$.



Question 6

A graph $y = f(x)$ has a minimum at $(3, -2)$. State the minimum point of $y = f(x - 1) + 4$.

Question 7

A graph $y = f(x)$ has a root at $x = 6$. State the corresponding root of $y = f(x + 2)$.

Question 8

Describe $y = 3(x + 2)^2 - 4$ as transformations of $y = x^2$, in order.

Question 9

A graph $y = f(x)$ has a root at $x = 8$. Find the root of $y = f(3x - 6)$ (factor first).

Question 10

A graph $y = 3f(x-2)+1$ has a minimum at $(5, 10)$. Find the corresponding minimum on $y = f(x)$.

Question 11 — Challenge

A graph $y = f(x)$ has a maximum at $(2, 5)$. State the coordinates of the corresponding point on $y = -2f(x-3)$.

Question 12 — Challenge

A city's rainfall model is $y = f(m)$ by month. A neighbouring city gets 20% more rain and its wet season arrives 1 month later. Write its model in terms of f.

Question 13 — Challenge

Given $f(x) = \sqrt{x}$, describe $y = f(9x)+2$ as a single vertical stretch plus a shift.

Question Recap & Answer Key

Question Recap

1. Describe the transformation from $y = x^2$ to $y = (x-3)^2 + 7$.
2. Describe $y = \frac{1}{2}x^2$ as a transformation of $y = x^2$.
3. Describe $y = \sqrt{-x}$ as a transformation of $y = \sqrt{x}$.
4. Describe $y = -x^2 - 5$ as two transformations of $y = x^2$.
5. Given $f(x) = x^2$, sketch and describe $y = f(2x)$.
6. A graph $y = f(x)$ has a minimum at $(3, -2)$. State the minimum point of $y = f(x-1) + 4$.
7. A graph $y = f(x)$ has a root at $x = 6$. State the corresponding root of $y = f(x+2)$.
8. Describe $y = 3(x+2)^2 - 4$ as transformations of $y = x^2$, in order.
9. A graph $y = f(x)$ has a root at $x = 8$. Find the root of $y = f(3x - 6)$ (factor first).
10. A graph $y = 3f(x-2) + 1$ has a minimum at $(5, 10)$. Find the corresponding minimum on $y = f(x)$.
11. A graph $y = f(x)$ has a maximum at $(2, 5)$. State the coordinates of the corresponding point on $y = -2f(x-3)$.
12. A city's rainfall model is $y = f(m)$ by month. A neighbouring city gets 20% more rain and its wet season arrives 1 month later. Write its model in terms of f.
13. Given $f(x) = \sqrt{x}$, describe $y = f(9x) + 2$ as a single vertical stretch plus a shift.

✓ Answer Key

1. right 3, up 7
2. vertical compression by $\frac{1}{2}$
3. reflection in the y-axis
4. reflected in the x-axis, then shifted down 5
5. $y = (2x)^2 = 4x^2$, horizontal compression by $\frac{1}{2}$
6. $(4, 2)$
7. $x = 4$
8. stretched vertically by 3, shifted left 2 and down 4; vertex $(-2, -4)$
9. compress horizontally by $\frac{1}{3}$ then shift right 2: root at $x = 8/3 + 2 = 14/3$
10. undo stretch/shift: $10 = 3y + 1 \Rightarrow y = 3$; undo horizontal shift: $x - 2 = 5 \Rightarrow x = 7$; point $(7, 3)$
11. Shift right 3: $x = 5$; reflect and stretch by 2: $y = -2(5) = -10$: point $(5, -10)$
12. $y = 1.2f(m-1)$
13. $y = \sqrt{9x} + 2 = 3\sqrt{x} + 2$: vertical stretch by 3, shifted up 2